Improved Synchronous Leader Election on an Anonymous Ring

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1 Introduction

We consider a directed ring of n anonymous processors: they run the same program, have no identifiers, and processor v_i can send messages only to v_{i+1} (indices modulo n). The network is synchronous, every processor knows n and has its own random number generator. The goal is to elect a leader, i.e. exactly one processor, so that at the end every processor knows whether it is the leader. We present the improved algorithm of Itai and Rodeh [1] (Section 2.4).

2 The algorithm

The algorithm proceeds in phases of n rounds, and all processors know a parameter q, $0 < q < n$. A *pebble* is a message without content; it travels around the ring and is dropped by its sender after exactly n rounds. In every phase:

1. each processor sends a pebble with probability q/n at the beginning of the phase;
2. each processor v records an n-bit word $W(v)$, where $W_i(v) = 1$ if and only if a pebble (possibly its own) passed through v in round i of the phase;
3. if all words $W(v)$ are distinct, the processor with the lexicographically smallest word becomes the leader; otherwise a new phase starts.

The algorithm relies on two invariants:

1. every pebble disappears at the end of its phase, so the words depend only on the current phase;
2. the words $W(v)$ are cyclic shifts of one word, since a pebble that passes v_j in round i passes v_{j+k} in round $(i + k) \bmod n$. Hence every processor can compute all the words by shifting its own.

3 Correctness

Theorem 1. *If the algorithm terminates, exactly one processor is the leader.*

Proof. At the end of a phase all processors see the same set of words, so either all of them start a new phase or all of them stop. They stop only when the words are distinct, and then the smallest word belongs to exactly one processor. □

Theorem 2. *The algorithm terminates with probability* 1.

Proof. If exactly one pebble is sent, all words contain a single 1 at different positions, so the phase succeeds. This happens with probability $q(1 - q/n)^{n-1} > 0$. Phases are independent, hence the probability of k consecutive failures tends to 0. □

4 Complexity

Theorem 3 (Itai, Rodeh, Theorem 2.2)**.** *As $n \to \infty$, the expected number of phases tends to $(1 - e^{-q})^{-1}$ and the expected number of bits sent per processor tends to $q(1 - e^{-q})^{-1}$.*

The paper states this theorem without proof; below is our proof.

Each processor sends a pebble independently with probability q/n, and the pebble of v_j passes v_0 in round $(n - j) \bmod n$. Hence $W = W(v_0)$ consists of n independent bits, each equal to 1 with probability q/n. A phase fails if and only if W equals its cyclic shift by some $1 \le k < n$; let P_n be the probability of this event.

Lemma 1. *If W equals its shift by some $1 \le k < n$, then for some divisor $r > 1$ of n the word W consists of r identical blocks.*

Proof. Let k be the smallest positive shift that does not change W. If k did not divide n, then $n = ak + b$ with $0 < b < k$; shifts by n and by ak do not change W, so neither does the shift by b, contradicting the choice of k. Thus W consists of $r = n/k > 1$ copies of a block of length k. □

Theorem 4. *For $n > q$*

$$\left(1 - \frac{q}{n}\right)^n \;\le\; P_n \;\le\; \left(1 - \frac{q}{n}\right)^n + \frac{2q^2}{\sqrt{n}}, \qquad \textit{hence} \qquad P_n \to e^{-q}.$$

Proof. The all-zero word, which occurs with probability $(1 - q/n)^n$, equals all its shifts; this gives the lower bound. If a failing word contains a 1, then by the lemma it consists of r identical blocks, so this 1 appears at the same position in all r blocks. For a fixed r there are n/r positions, each giving r ones with probability $(q/n)^r$, so by the union bound this case has probability at most $\frac{n}{r}(q/n)^r \le q^2/n$. Since n has at most $2\sqrt{n}$ divisors, summing over r gives at most $2q^2/\sqrt{n}$. Both bounds tend to e^{-q}. □

Proof of Theorem 2.2. Phases are independent and each succeeds with probability $1 - P_n$, so the number of phases is geometric with expectation $1/(1 - P_n) \to 1/(1 - e^{-q})$. In every phase n processors send a pebble with probability q/n each, so on average q pebbles are sent. Each pebble makes n steps, i.e. costs one bit per processor. Hence the expected number of bits per processor is

$$q \cdot \mathbb{E}[\text{number of phases}] \;=\; \frac{q}{1 - P_n} \;\longrightarrow\; \frac{q}{1 - e^{-q}}.$$ □

References

[1] A. Itai, M. Rodeh, *Symmetry breaking in distributed networks*, Information and Computation 88(1), 60–87 (1990).